

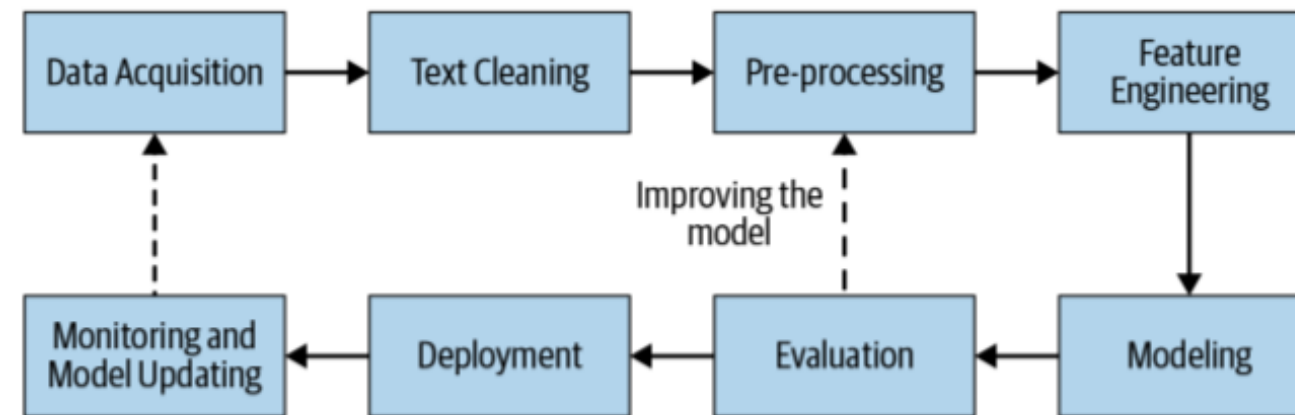
Applied NLP

DATA 641

Lecture 2 — NLP Pipeline; Data Acquisition

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Generic NLP Pipeline



Main components of an NLP pipeline

- **Data Acquisition:** Collect the data relevant to the given task
- **Text Cleaning:** Data needs to be cleaned
- **Pre-processing:** Convert text data into a canonical form
- **Feature Engineering:** Make data understandable by algorithms
- **Modeling/Evaluation:** Built and compare models
- **Deploy model in production**
- **Update model to keep up model performance**

Data Acquisition

Data is the heart of any ML system. In most industrial projects, it is often the data that becomes the bottleneck.

Example Let's say we are asked to develop an NLP system to identify whether an incoming customer query (for example, using a chat interface) is a sales inquiry or a customer care inquiry. Depending on the type of query, it should be automatically routed to the right team. How can one go about building such a system? Well, the answer depends on the type and amount of data we have to work with.

In an ideal setting, we will have the required datasets with thousands or millions of data points. In such cases, we do not have to worry about data acquisition.

However, in many real-world projects, one is not so lucky. Let us look at what we can do in a less-than-ideal scenario. How can we get such data?

- **Public datasets:** [Use a public dataset](#). We could see if there are any public datasets available that we can leverage. If you find a suitable dataset that is similar to the task at hand, great! Build a model and evaluate. If not, then what?
- **Scrape data:** Scrape data from public websites and get it labeled by human annotators. For many industrial settings, gathering data from external sources does not suffice because the data does not contain nuances like product names or product specific user behavior and thus might be very different from the data seen in production environments.
- **Public intervention:** Developed mostly to serve users via a feature or product. (Google; Facebook; Netflix are known for this). Product intervention is often the best way to collect data for building intelligent applications in industrial settings.

- **Data augmentation:** Start with a small dataset and use some tricks to create more data
 - **Synonym replacement:** Randomly select words in a sentence and replace them by synonyms
 - **Back translation:** Translate sentence S1 from English to Greek obtain sentence S2, then take S2 and translate it to English and obtain sentence S3. If S1 and S3 are different add S3 to the dataset
 - **Bigram flipping:** Take a bigram "going to" and replace it with the flipped version "to going"
 - **Replacing entities:** For example, in "I live in California", replace "California" with "London"
 - **Adding noise to the data:** Add a bit of noise to the data to train robust models. For example, choose a word in a sentence and replace it with another word that's closer in spelling to the first word
 - **Advanced Techniques:** Active Learning, Easy data augmentation, [Snorkel](#).